

- Concentration inequalities: Inequalities that bound the probability of a random variable deviating from some value.

$$|X - a| \leq \epsilon \text{ w.p. } 1 - \delta$$

random variable X , some value a , ϵ is small

Example: Markov's inequality, Chebyshev's inequality, Chernoff bound
Hoeffding's inequality, Azuma's inequality

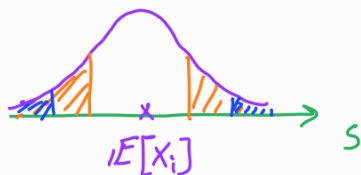
Recall

$X_i \sim D$ with mean μ and variance σ^2 , e.g., $\mathcal{N}(\mu, \sigma^2)$

$$S = \frac{1}{n} \sum_{i=1}^n X_i$$

random variable $f(X_1, X_2, \dots, X_n)$, i.i.d.

random variable $f(X_1, X_2, \dots, X_n)$



LLN as $n \rightarrow \infty \Rightarrow S \rightarrow \mathbb{E}[X_i] = \mu$

CLT as $n \rightarrow \infty \Rightarrow S \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$

Hoeffding's inequality: Suppose $X_1, X_2, \dots, X_t$ are independent random variables such that

$$\begin{cases} \forall i : a_i \leq X_i \leq b_i & \text{almost surely} \\ \text{and} & S_t = \sum_{i=1}^t X_i \end{cases}$$

Then, the following holds:

$$\mathbb{P}[|S_t - \mathbb{E}[S_t]| \geq \epsilon] \leq 2 \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^t (b_i - a_i)^2}\right) \quad \forall \epsilon \geq 0.$$

$$\bar{S}_t = \frac{1}{t} \sum_{i=1}^t X_i \quad a \leq X_i \leq b \quad \longrightarrow \quad \mathbb{P}[|\bar{S}_t - \mathbb{E}[\bar{S}_t]| \geq \epsilon] \leq 2 \exp\left(\frac{-2\epsilon^2 t}{|b-a|^2}\right) \quad \forall \epsilon \geq 0$$

- **Stochastic Process:** A collection of random variables, indexed by a set, e.g., time or space, defined over a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

$$X_0, X_1, X_2, \dots$$

$$(X_t)_{t=0}^{\infty} \quad \text{or} \quad \{X_t : t \in \mathbb{N}_0\}$$

Example: Bernoulli process, Gaussian process, random walk, Wiener process, Poisson process

- **Markov process:** A class of stochastic processes, discrete or continuous, that follow Markov property.

Markov property: Given the present, the future is independent of the past.

$$\mathbb{P}[X_{t+1} \mid X_0, X_1, X_2, \dots, X_t] = \mathbb{P}[X_{t+1} \mid X_t] \quad \forall t \in \mathbb{N}_0$$

discrete-time
Markov process

$$\mathbb{P}[X_{t+1}, X_{t+2} \mid X_0, X_1, \dots, X_t] = \mathbb{P}[X_{t+1}, X_{t+2} \mid X_t]$$

$$\mathbb{P}[X_{t+1} \mid X_0, X_1, \dots, X_{t-1}] = \mathbb{P}[X_{t+1} \mid X_{t-1}]$$

Type $\begin{cases} \rightarrow \text{discrete time} \\ \rightarrow \text{continuous time} \end{cases}$

Type $\begin{cases} \rightarrow \text{discrete space} \\ \rightarrow \text{continuous space} \end{cases}$

$\begin{cases} \text{time-homogeneous} : \mathbb{P}[X_{t+1} | X_t] = \mathbb{P}[X_{t+m+1} | X_{t+m}] \quad \forall m \\ \text{time-inhomogeneous} \end{cases}$

- Martingale process: A class of stochastic processes such that

$$\begin{cases} \mathbb{E}[|X_t|] < \infty \\ \mathbb{E}[X_t | X_0, X_1, \dots, X_{t-1}] = X_{t-1} \end{cases} \quad \forall t \in \{0, 1, 2, \dots\} .$$

Azuma's inequality: suppose $\{X_t : t \in \mathbb{N}_0\}$ is a martingale process such that

$$\forall t \in \mathbb{N}: |X_t - X_{t-1}| \leq c_t \quad \text{almost surely} .$$

Then, the following holds:

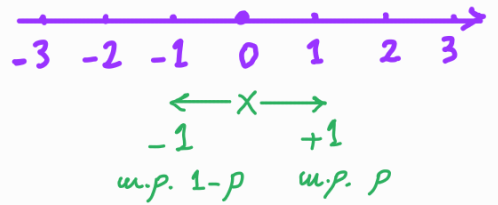
$$\mathbb{P}[|X_T - X_0| \geq \varepsilon] \leq 2 \exp \left(\frac{-\varepsilon^2}{2 \sum_{t=1}^T c_t^2} \right) \quad \forall \varepsilon \geq 0 .$$

Example: Random walk

X RV for steps

$$\Omega_X = \{+1, -1\} \quad \mathcal{F}_X = 2^{\Omega_X} \quad p \in (0, 1)$$

$$(X_t)_{t=0}^{\infty} \text{ SP where } X_t = \begin{cases} +1 & \text{w.p. } p \\ -1 & \text{w.p. } 1-p \end{cases}$$



$$\mathbb{P}[X_{t+1} | X_0, X_1, \dots, X_t] = \mathbb{P}[X_{t+1} | X_t] = \mathbb{P}[X_{t+1}]$$

Markov

independent

$X_t, X_{t'} \quad \forall t, t' \text{ where } t \neq t'$

$$\mathbb{E}[X_t | X_0, X_1, \dots, X_{t-1}] = \mathbb{E}[X_t] = p - (1-p) = 2p - 1$$

$$p \in (0, 1) \leftarrow \neq X_{t-1}$$

not martingale

S RV for location

$$\Omega_S = \mathbb{Z} \quad \mathcal{F}_S = 2^{\Omega_S}$$

$$(S_t)_{t=0}^{\infty} \text{ SP where } S_0 = 0 \text{ and } S_t = \sum_{i=1}^t X_i \text{ for } t \geq 1$$

not independent $S_t, S_{t'} \quad \forall t, t' \text{ where } t \neq t'$

$$\mathbb{P}[S_{t+1} | S_0, S_1, \dots, S_t] = \mathbb{P}[S_{t+1} | S_t]$$

Markov

$$\mathbb{E}[S_t | S_0, S_1, \dots, S_{t-1}] = \mathbb{E}[S_t | S_{t-1}]$$

$$= p(S_{t-1} + 1) + (1-p)(S_{t-1} - 1)$$

$$= S_{t-1} + 2p - 1$$

if $p = 0.5 \rightarrow$ martingale ; otherwise, not martingale